Predatory Trading

Replicating, verifying and extending Brunnermeier & Pedersen (2005)

Harsh Gupta (22IM3FP29)

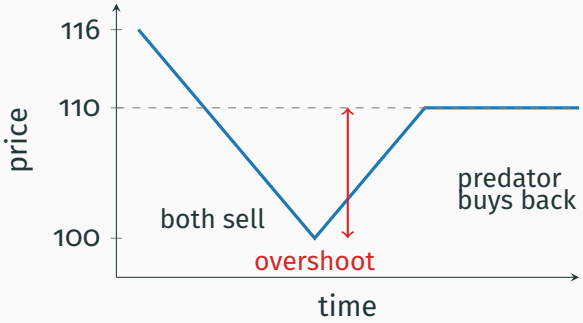
Market Microstructure · Prof. Ajay Kumar Mishra
Indian Institute of Technology Kharagpur

Why this matters

> *"…if lenders know that a hedge fund needs to sell something quickly, they will sell the same asset — driving the price down even faster. Goldman, Sachs & Co. and other counterparties to LTCM did exactly that in 1998."*
>
> — *Business Week*, 2001 (quoted in the paper, Table I)

- 1987 crash: traders sold *ahead of* portfolio insurers (Brady Report)
- Fed on LTCM: unwinding could cause "large losses, or worse" for others

The idea in one picture



- A large trader is **forced to sell**
- Rivals know it — and **sell too**
- Price falls *below* its new level
- Rivals **buy back** at the bottom

Victim's extra loss = predators' profit.
Liquidity vanishes *when it is needed most*.

The model

Ingredient	Meaning	Formula
Demand curve	selling to ordinary investors lowers the price	$p(t) = \mu - \lambda\,(S - X(t))$
Limited capital	each big trader holds at most $\bar{x}$	$\lvert x_i \rvert \le \bar{x}$
Speed limit	big traders together trade $\le A$ per unit time	$\left\lvert \sum_i a_i \right\rvert \le A$
Distress at t_0	I_d traders must sell at speed A/I	$I_p = I - I_d$ predators

- X = total holdings of the big traders; λ = price impact
- Everyone knows who is distressed; predators maximise expected profit
- Paper's example: $S = 40$, $\mu = 140$, $\lambda = 1$, $A = 20$, $\bar{x} = 10$, $x(t_0) = 8$

The key insight (Lemma 1)

A predator's problem reduces to

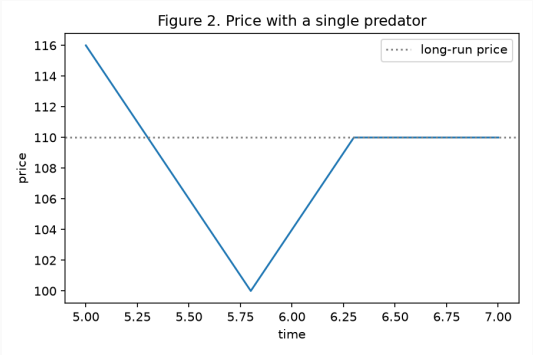
$$\min_{a_i} \int_0^T a_i(t)\, X_{-i}(t)\, dt \quad \text{s.t. } x_i(T) = \bar{x}$$

where X_{-i} is **everyone else's** holdings.

- Its *own* price impact cancels out (paid when buying, earned back when selling)
- So: **sell while X_{-i} is falling, buy once it stops**
- Profit comes from the *victim's* price impact $\Rightarrow$ predation, not manipulation

For us: the objective is *linear* in the predator's trades $\Rightarrow$ best responses are linear programs (slide 8).

One predator (Prop. 1): our simulation



- $t = 5 \to 5.8$: both sell 8 shares; 16 sold $\Rightarrow$ 116 $\to$ **100**
- $t = 5.8 \to 6.3$: predator buys back to 10 $\Rightarrow$ **110**
- Overshoot of **10**

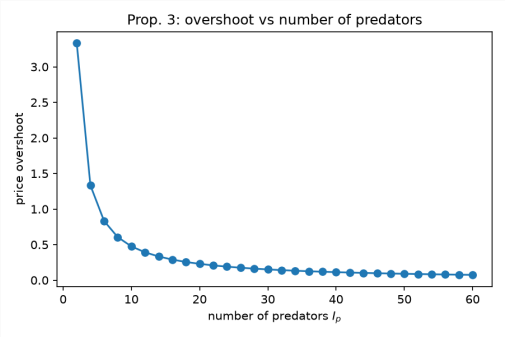
Why sell below 110? The last share sells at $\approx$ 101 and is bought back at $\approx$ 100.5: **+0.50**. Our simulation reproduces this to the cent.

Live demo

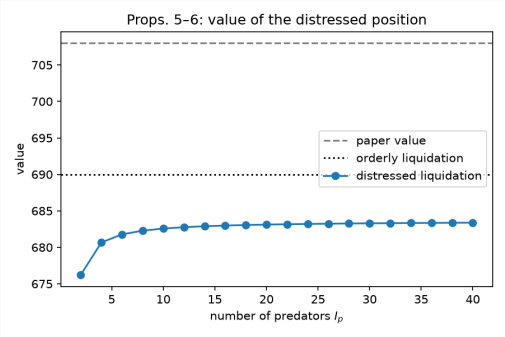
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uv run streamlit run app.py
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“Single predator” $\rightarrow$ “Many predators”

Competition, and what a position is worth

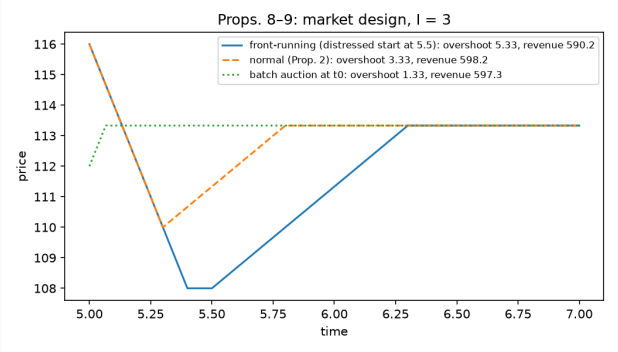


Prop. 3: predators race to rebuild first $\Rightarrow$ overshoot $3.33 \to 0.83 \to 0.26$ for $2, 6, 18$ predators



Props. 5–6: paper $>$ orderly $>$ distressed value (e.g. $928 > 896 > 864$), even with many predators

Market design (Props. 8–9)



Market	Overshoot	Seller gets
Front-running	5.33	590.2
Normal	3.33	598.2
Batch auction	1.33	597.3

Our finding: a batch auction (circuit breaker) calms the price, but does **not** raise the seller's revenue — everything sells at one price.

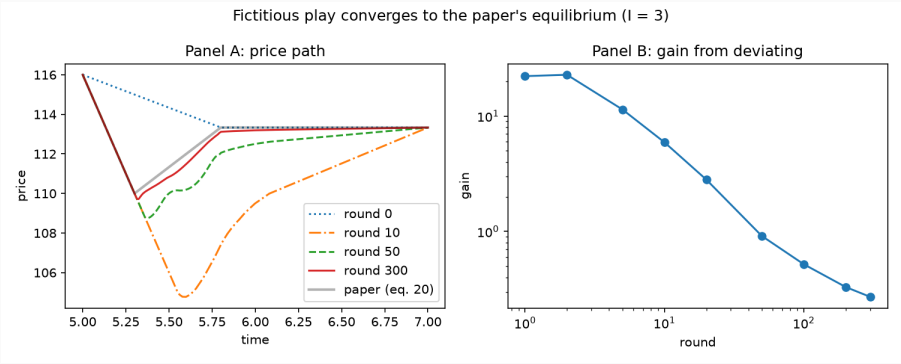
Does it hold up? (1) Best response as a linear program

Fix everyone else at the paper's strategy; let a solver find one predator's *best possible* strategy.

Case	Paper's strategy pays	Best possible pays
Prop. 1 (one predator)	1214.000000	1214.000000
Prop. 2, $l = 3$	787.111111	787.111111
Prop. 2, $l = 9$	261.444444	261.444444
Prop. 2′ (spare capital)	702.000000	702.000000
Prop. 8 (front-running)	791.111111	791.111111

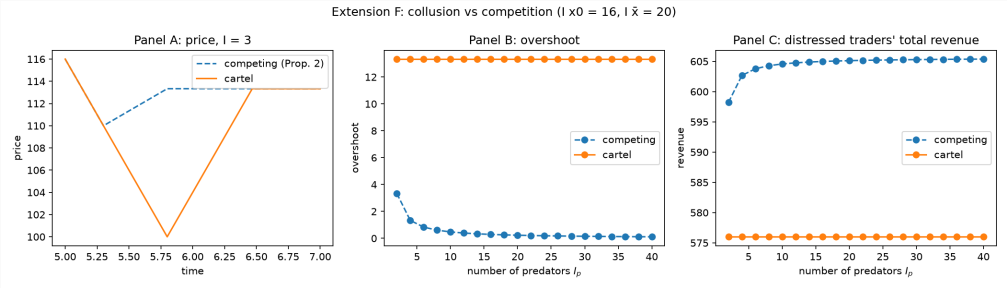
No profitable deviation $\Rightarrow$ the paper's strategies are equilibria. Sensitivity check: without the speed limit the solver earns 1324, so the test can fail.

Does it hold up? (2) Traders learn to prey



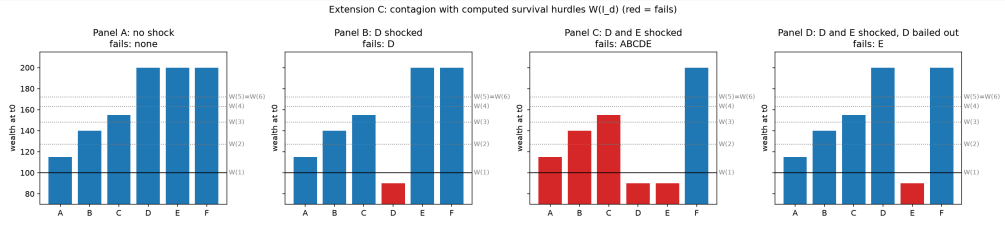
Start: predators just buy (no predation). Each round they move a step toward their best response (fictitious play). Round 10: they **overdo it** (price $\approx$ 105). Round 300: the paper's path. Gain from deviating 22 $\rightarrow$ 0.27. Predation is not assumed — it is learned.

Extension: collusion



- A cartel acts as one big predator: overshoot **13.33** vs 3.33, and it does not shrink with size
- A *pure transfer*: predators gain exactly what the victims lose
- Unstable: a member gains 16.67 by cheating (2.1% of its profit; 4.3% with 6 members)

Extension: contagion

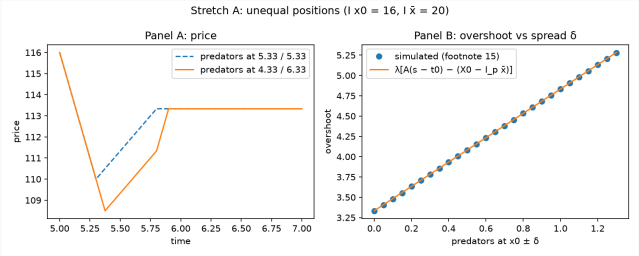


- We *computed* the paper's survival hurdles $W(I_d)$ (eq. 21) by sequential linear programming: $W = 100, 127, 148, 163, 172, 172$
- Best way to survive: sell, then **buy at the bottom** to prop up your own paper wealth
- D and E shocked $\Rightarrow$ ripple **D, E** $\rightarrow$ **A** $\rightarrow$ **B** $\rightarrow$ **C**; bailing out D stops it

Live demo

“Contagion” page: shock D and E, then bail out D

Extension: predators with unequal positions



Paper (footnote 15, no proof): more dispersed holdings $\Rightarrow$ more overshoot.

We confirm the footnote's equilibrium with the solver, and sharpen the claim:

$$\text{overshoot} = \lambda\,[A(s - t_0) - (X_0 - I_p\bar{x})]$$

s depends only on the **smallest** predator: it keeps selling longest, and everyone sells with it.

Takeaways

Economics

- Liquidity disappears when it is needed most
- Secrecy protects funds (disclosure invites predation)
- Circuit breakers calm prices; bailouts stop cascades
- Collusion makes predation far worse — but is unstable

What we added

- Exact simulator reproducing Props. 1–9 and Figs. 1–3
- Independent check: no profitable deviation (LP), and traders *learn* the equilibrium
- New results on collusion, contagion and unequal positions
- Interactive app for every result

115 automated tests · `github.com/dudegladiator/predatory-trading`

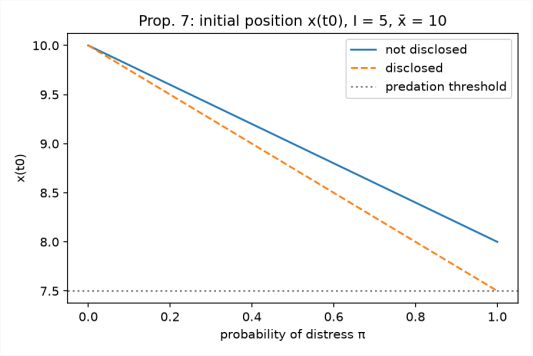
Thank you

Questions?

Backup: spare capital and disclosure (Props. 2′, 7)

Prop. 2′: if $x(t_0) < \frac{I_p - 1}{I - 1}\,\bar{x}$, competition to buy beats the urge to prey: predators buy at once, no overshoot.

Prop. 7: traders build positions large enough that predation happens with positive probability. Disclosure shrinks positions ($9.0 \to 8.75$ for $I = 5$, $\pi = 0.5$) — but never below the predation threshold.



Backup: how the simulation stays exact

- Time grid $t_0, t_0 + dt, \ldots, T$; trading rates constant within a step
- Price is linear in holdings $\Rightarrow$ linear within a step $\Rightarrow$ cash $= -\sum a_k \, dt \, (p_k + p_{k+1})/2$ is **exact**
- Switch times between grid points: rates averaged over the step
- Trading faster than the speed caps (eqs. 8–10) is charged $\gamma = \lambda I \bar{x}$ per excess share — added after we caught averaged learning strategies breaking the limit

Backup: computing the survival hurdle

$$W(I_d) = \underline{W} + D(I_d), \qquad D(I_d) = -\max_{a_i} \min_{t} \Delta W_i(t)$$

$$\Delta W_i(t) = \lambda \left[\tfrac{1}{2}(x_t^2 - x_0^2) + \textstyle\sum_{k<t} \bar{x}_k \, \Delta X_{-i,k} \right]$$

- x^2 makes max–min non-convex $\Rightarrow$ replace $\frac{1}{2}x^2$ by its tangent (never above it), solve the LP, re-linearise, repeat
- Several starting paths; 25 random starts all reach the same values
- Identical at $dt = 0.005, 0.0025, 0.001 \Rightarrow$ no grid artifact